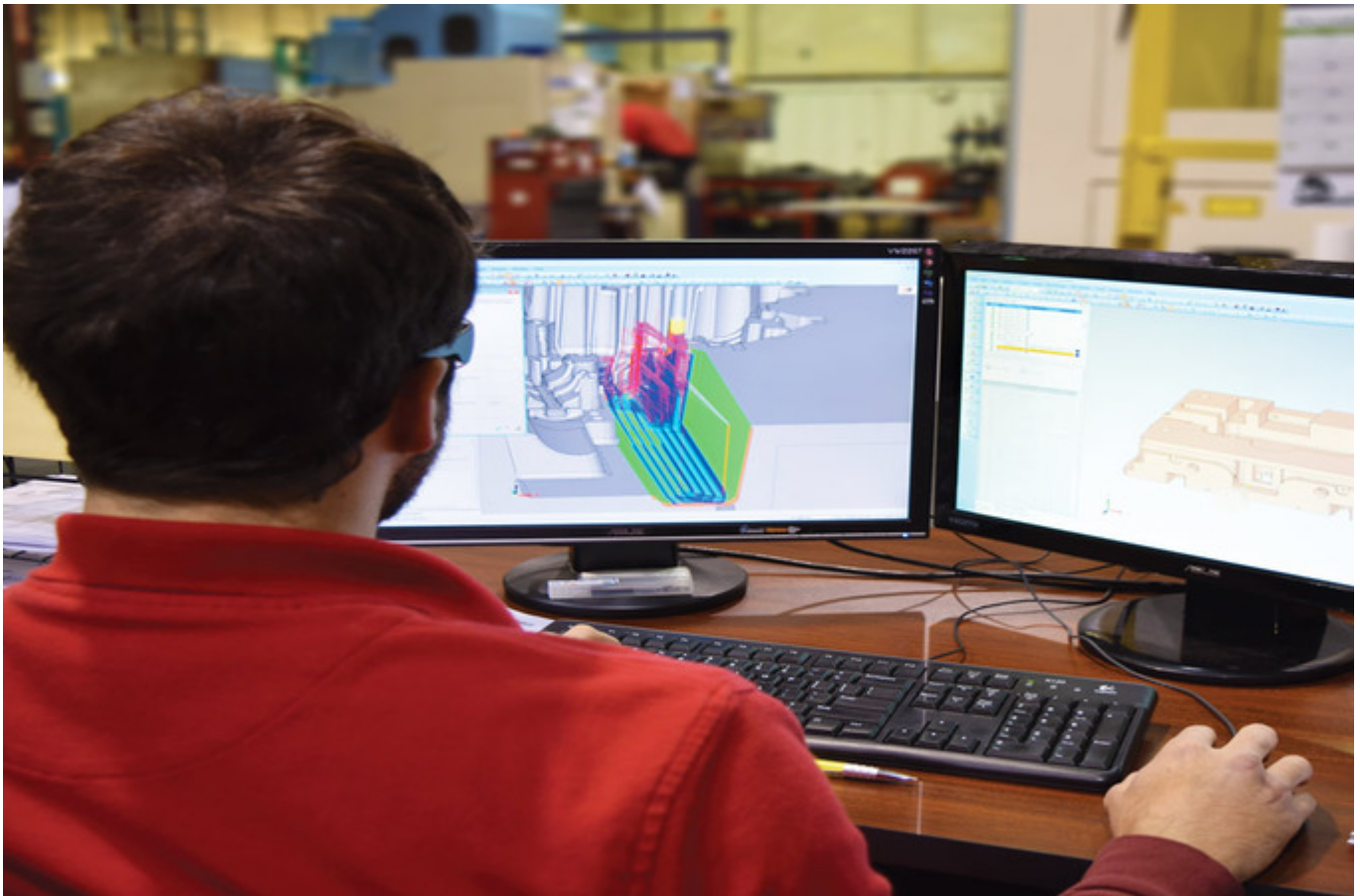


Qualification Pack



Jr. Designer - Plastic Products

QP Code: RSC/Q8003

Version: 3.0

NSQF Level: 5

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket
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Qualification Pack

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RSC/Q8003: Jr. Designer - Plastic Products

Brief Job Description

The Jr. Designer - Plastic Products supports the design and development of plastic components by assisting in conceptualization, CAD modeling, and prototyping. The role involves creating and modifying 2D/3D drawings, collaborating with senior designers and engineers, and participating in product validation and testing. The designer ensures functional, manufacturable, and customer-oriented product designs in alignment with industry standards.

Personal Attributes

The individual at work should be creative and innovative with an eye to detail.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N8003: Assist in product design & development](#)
2. [RSC/N4409: Create and Modify 2D/3D Drawings using CAD Software](#)
3. [RSC/N4410: Assist in Prototyping and Product Testing](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Non - Tyre
Occupation	Plastic Product Designing
Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0201

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Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (Or Equivalent) OR 12th grade Pass with 3 Years of experience Relevant OR 10th grade pass with 6 Years of experience Relevant OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience Relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Basic knowledge of Designing
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	3.0
Reference code on NQR	QG-05-RI-04028-2025-V2-RSDC
NQR Version	2

Remarks:

TBD

Qualification Pack

RSC/N8003: Assist in product design & development

Description

This unit covers the process of supporting the design and development of plastic products, from understanding design briefs to preparing basic concept sketches and assisting in feasibility analysis.

Scope

The scope covers the following :

- Understand design brief and specifications
- Conduct product benchmarking and research
- Create design concepts
- Assist in creating functional mock-ups
- Support design feasibility assessment
- Maintain project records
- Maintain records and follow-up

Elements and Performance Criteria

Understand design brief and specifications

To be competent, the user/individual on the job must be able to:

- PC1.** identify product requirements based on client/supervisor inputs
- PC2.** identify the functionality and aesthetic needs of the product
- PC3.** assess the plastic material constraints related to the design
- PC4.** consider environmental, economic, and regulatory factors influencing design

Conduct research and benchmarking

To be competent, the user/individual on the job must be able to:

- PC5.** collect reference data and analyze similar existing products
- PC6.** study functional, aesthetic, and ergonomic parameters
- PC7.** document benchmark metrics and identify innovation opportunities

Prepare basic concept sketches

To be competent, the user/individual on the job must be able to:

- PC8.** generate freehand sketches or concept diagrams
- PC9.** propose variations and alternatives in initial design
- PC10.** incorporate supervisor feedback into revised concepts

Assist in product detailing

To be competent, the user/individual on the job must be able to:

- PC11.** support the preparation of dimensional sketches
- PC12.** help identify tolerances and draft angles based on process constraints
- PC13.** suggest material thickness based on load and stress considerations

Support feasibility analysis

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To be competent, the user/individual on the job must be able to:

PC14. assist in checking manufacturability and tooling aspects

PC15. participate in material and cost feasibility studies

PC16. apply sustainability and recycling considerations during selection

Contribute to cross-functional review

To be competent, the user/individual on the job must be able to:

PC17. coordinate with marketing, manufacturing, and testing teams

PC18. document discussions and feedback during internal reviews

PC19. revise design data based on team inputs

Maintain records and follow-up

To be competent, the user/individual on the job must be able to:

PC20. maintain version history of design sketches and data

PC21. ensure confidentiality and handle proprietary information carefully

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. key stages in plastic product development lifecycle

KU2. basic principles of product design and industrial design

KU3. types of plastic materials and their influence on design features

KU4. difference between functional and decorative components

KU5. ergonomic factors in plastic product design

KU6. methods for performing benchmarking and competitor analysis

KU7. importance of intellectual property and design patents

KU8. common manufacturing constraints affecting design

KU9. basics of cost estimation for design alternatives

KU10. guidelines for eco-design and sustainable plastic product development

KU11. material selection criteria including load, impact, and temperature resistance

KU12. limitations and strengths of various plastic forming processes

KU13. use of hand sketching and digital sketching tools

KU14. purpose and structure of design documentation

KU15. tolerance implications on design and assembly

KU16. compliance requirements such as RoHS, REACH, or BIS

KU17. basics of design for manufacturability and assembly (DFMA)

KU18. role of aesthetics and form development in consumer plastics

KU19. implications of product failure and how to mitigate risks during design

KU20. protocols for internal review and documentation during development

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** read and interpret basic product specifications and design briefs
- GS2.** write notes and observations clearly while researching product benchmarks
- GS3.** communicate effectively with senior designers and cross-functional team members
- GS4.** apply basic arithmetic and geometry for concept development
- GS5.** use digital tools like MS PowerPoint or Google Slides for concept presentations
- GS6.** plan and organize research and sketching activities efficiently
- GS7.** take responsibility for assigned design-related support tasks
- GS8.** think creatively to suggest alternative product forms
- GS9.** identify trends and needs in the plastic product market through research
- GS10.** maintain accuracy while documenting concept versions and feedback

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand design brief and specifications</i>	3	13	-	-
PC1. identify product requirements based on client/supervisor inputs	1	3	-	-
PC2. identify the functionality and aesthetic needs of the product	1	3	-	-
PC3. assess the plastic material constraints related to the design	1	3	-	-
PC4. consider environmental, economic, and regulatory factors influencing design	1	4	-	-
<i>Conduct research and benchmarking</i>	5	11	-	-
PC5. collect reference data and analyze similar existing products	1	3	-	-
PC6. study functional, aesthetic, and ergonomic parameters	2	4	-	-
PC7. document benchmark metrics and identify innovation opportunities	2	4	-	-
<i>Prepare basic concept sketches</i>	4	10	-	-
PC8. generate freehand sketches or concept diagrams	1	3	-	-
PC9. propose variations and alternatives in initial design	2	4	-	-
PC10. incorporate supervisor feedback into revised concepts	1	3	-	-
<i>Assist in product detailing</i>	6	12	-	-
PC11. support the preparation of dimensional sketches	2	4	-	-
PC12. help identify tolerances and draft angles based on process constraints	2	4	-	-
PC13. suggest material thickness based on load and stress considerations	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Support feasibility analysis</i>	6	12	-	-
PC14. assist in checking manufacturability and tooling aspects	2	4	-	-
PC15. participate in material and cost feasibility studies	2	4	-	-
PC16. apply sustainability and recycling considerations during selection	2	4	-	-
<i>Contribute to cross-functional review</i>	4	8	-	-
PC17. coordinate with marketing, manufacturing, and testing teams	1	2	-	-
PC18. document discussions and feedback during internal reviews	1	2	-	-
PC19. revise design data based on team inputs	2	4	-	-
<i>Maintain records and follow-up</i>	2	4	-	-
PC20. maintain version history of design sketches and data	1	2	-	-
PC21. ensure confidentiality and handle proprietary information carefully	1	2	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N8003
NOS Name	Assist in product design & development
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Production/Manufacturing Plastic Processing
NSQF Level	5
Credits	4
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N4409: Create and Modify 2D/3D Drawings using CAD Software

Description

This unit covers creating, editing, and managing 2D/3D CAD models of plastic parts and assemblies, including dimensioning, tolerancing, and rendering using standard CAD software tools.

Scope

The scope covers the following :

- Prepare 2D Drawings
- Model 3D Parts
- Assemble Parts Virtually
- Modify CAD Models
- Prepare files for sharing and documentation
- Ensure Quality and Compliance of Drawings

Elements and Performance Criteria

Prepare 2D Drawings

To be competent, the user/individual on the job must be able to:

- PC1.** create orthographic, sectional, and auxiliary views
- PC2.** add dimensional tolerances, surface finish, and material details
- PC3.** follow BIS/ISO standards while creating drawing templates
- PC4.** generate exploded views and part lists for assemblies

Model 3D Parts

To be competent, the user/individual on the job must be able to:

- PC5.** use solid modeling tools to create parts using extrude, revolve, and loft features
- PC6.** apply fillets, chamfers, ribs, and cuts as needed for strength and design intent
- PC7.** use parametric constraints to ensure model accuracy

Assemble parts virtually

To be competent, the user/individual on the job must be able to:

- PC8.** apply assembly mates, joints, and alignments
- PC9.** check and resolve part interferences and motion constraints
- PC10.** animate assemblies to visualize movement and alignment

Modify CAD Models

To be competent, the user/individual on the job must be able to:

- PC11.** update parts and assemblies based on feedback
- PC12.** revise dimensions, features, or constraints as required
- PC13.** maintain version control of design files

Prepare files for sharing and documentation

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC14.** export files in required formats (IGES, STEP, STL)
- PC15.** compile drawings and models into folders with proper nomenclature
- PC16.** generate BOM and associated data reports

Ensure quality and compliance of drawings

To be competent, the user/individual on the job must be able to:

- PC17.** review drawings for missing data or errors
- PC18.** apply drawing verification checklists before final submission
- PC19.** follow file storage and backup protocols
- PC20.** communicate drawing changes to design and prototyping teams

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of CAD software commonly used in plastic design (AutoCAD, SolidWorks, etc.)
- KU2.** basics of orthographic projection and engineering drawing standards
- KU3.** three-D modeling features and techniques for plastic parts
- KU4.** types and applications of assembly constraints
- KU5.** parametric modeling vs direct modeling
- KU6.** dimensioning rules and tolerance stacking
- KU7.** geometric dimensioning and tolerancing (GD&T) symbols and interpretation
- KU8.** use of CAD layers, blocks, and templates
- KU9.** file management protocols in design projects
- KU10.** steps to animate and simulate motion in assemblies
- KU11.** standards for plastic part documentation (e.g., BIS, ISO)
- KU12.** CAD system settings for file exports and formats
- KU13.** version control and model revision best practices
- KU14.** material and finish annotations for plastic components
- KU15.** interaction of CAD with downstream tools like CAM/3D printers
- KU16.** best practices for layout, clarity, and readability in technical drawings
- KU17.** identification and troubleshooting of modeling errors
- KU18.** significance of Bill of Materials (BOM)
- KU19.** integration of external references and modular assemblies
- KU20.** data security and backup procedures for design files

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret technical sketches and translate into digital models
- GS2.** document changes made to CAD files systematically
- GS3.** communicate with team members about design modifications

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- GS4.** apply mathematical skills for scaling, proportions, and tolerances
- GS5.** operate CAD tools proficiently for drawing and modeling
- GS6.** plan tasks based on design milestones and priorities
- GS7.** troubleshoot and resolve software-related modeling issues
- GS8.** focus on detail while verifying dimensional and geometric accuracy
- GS9.** manage data backup and folder hierarchy independently
- GS10.** stay updated with new CAD techniques and features through self-learning

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare 2D Drawings</i>	12	12	-	-
PC1. create orthographic, sectional, and auxiliary views	3	3	-	-
PC2. add dimensional tolerances, surface finish, and material details	3	3	-	-
PC3. follow BIS/ISO standards while creating drawing templates	3	3	-	-
PC4. generate exploded views and part lists for assemblies	3	3	-	-
<i>Model 3D Parts</i>	6	9	-	-
PC5. use solid modeling tools to create parts using extrude, revolve, and loft features	2	3	-	-
PC6. apply fillets, chamfers, ribs, and cuts as needed for strength and design intent	2	3	-	-
PC7. use parametric constraints to ensure model accuracy	2	3	-	-
<i>Assemble parts virtually</i>	4	9	-	-
PC8. apply assembly mates, joints, and alignments	2	3	-	-
PC9. check and resolve part interferences and motion constraints	2	3	-	-
PC10. animate assemblies to visualize movement and alignment	2	3	-	-
<i>Modify CAD Models</i>	4	9	-	-
PC11. update parts and assemblies based on feedback	2	3	-	-
PC12. revise dimensions, features, or constraints as required	2	3	-	-
PC13. maintain version control of design files	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare files for sharing and documentation</i>	6	9	-	-
PC14. export files in required formats (IGES, STEP, STL)	2	3	-	-
PC15. compile drawings and models into folders with proper nomenclature	2	3	-	-
PC16. generate BOM and associated data reports	2	3	-	-
<i>Ensure quality and compliance of drawings</i>	8	12	-	-
PC17. review drawings for missing data or errors	2	3	-	-
PC18. apply drawing verification checklists before final submission	2	3	-	-
PC19. follow file storage and backup protocols	2	3	-	-
PC20. communicate drawing changes to design and prototyping teams	2	3	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4409
NOS Name	Create and Modify 2D/3D Drawings using CAD Software
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Design
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N4410: Assist in Prototyping and Product Testing

Description

This unit deals with assisting in the creation of prototypes using 3D printing, CNC, or other methods and participating in the testing and evaluation of plastic products.

Scope

The scope covers the following :

- Prepare Files and Materials for Prototyping
- Assist in Prototype Fabrication and Validation
- Assist in Functional Testing
- Support Iteration and Documentation

Elements and Performance Criteria

Prepare Files and Materials for Prototyping

To be competent, the user/individual on the job must be able to:

- PC1.** convert CAD models into STL/STEP formats
- PC2.** verify model geometry and orientation for prototyping
- PC3.** coordinate with service providers or in-house labs

Assist in Prototype Fabrication and Validation

To be competent, the user/individual on the job must be able to:

- PC4.** support 3D printing, CNC machining, or vacuum casting processes
- PC5.** inspect physical output for surface quality and dimensional accuracy
- PC6.** perform basic post-processing like trimming or assembly
- PC7.** participate in fitment trials or ergonomic checks
- PC8.** report observed deviations and provide input for design iteration

Assist in Functional Testing

To be competent, the user/individual on the job must be able to:

- PC9.** measure strength, stability, or deformation under stress
- PC10.** collect and organize testing data
- PC11.** identify visible or functional defects in prototypes

Support Iteration and Documentation

To be competent, the user/individual on the job must be able to:

- PC12.** incorporate design changes based on test results
- PC13.** track design revisions and create test summaries
- PC14.** follow documentation formats for prototype evaluation
- PC15.** assist in preparing technical review reports

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** basic prototyping technologies like FDM, SLA, CNC, vacuum casting, etc.
- KU2.** process parameters and materials for prototype making
- KU3.** STL file generation and slicing software
- KU4.** fit and tolerance considerations in prototypes
- KU5.** steps involved in prototype post-processing
- KU6.** standard performance and durability tests for plastics
- KU7.** common types of failure in plastic prototypes
- KU8.** visual inspection and surface finish evaluation
- KU9.** data recording and basic statistical analysis
- KU10.** iterative nature of design-prototype-test cycles
- KU11.** role of prototyping in validating product usability
- KU12.** compliance standards relevant to prototype evaluation
- KU13.** layout and format of prototype testing reports
- KU14.** communication protocols with vendors and testing labs
- KU15.** safety considerations during prototype handling
- KU16.** timelines and cost factors in prototyping
- KU17.** measuring instruments and interpretation of results
- KU18.** techniques to improve the prototype-to-product transition
- KU19.** impact of prototyping feedback on product finalization
- KU20.** importance of maintaining confidentiality during early testing

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow SOPs for prototyping tools and equipment
- GS2.** write accurate notes and reports for test observations
- GS3.** interact with vendors and supervisors regarding prototype specifications
- GS4.** apply basic math for measuring, comparing, and analyzing test data
- GS5.** operate basic tools and devices during prototype inspection
- GS6.** organize files, reports, and feedback systematically
- GS7.** take initiative to validate design assumptions through testing
- GS8.** respond promptly to feedback and iterate designs accordingly
- GS9.** maintain attention to detail while performing inspection or trials
- GS10.** uphold safety and cleanliness in lab/workshop environments

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare Files and Materials for Prototyping</i>	7	14	-	-
PC1. convert CAD models into STL/STEP formats	3	5	-	-
PC2. verify model geometry and orientation for prototyping	2	4	-	-
PC3. coordinate with service providers or in-house labs	2	5	-	-
<i>Assist in Prototype Fabrication and Validation</i>	13	23	-	-
PC4. support 3D printing, CNC machining, or vacuum casting processes	3	5	-	-
PC5. inspect physical output for surface quality and dimensional accuracy	2	4	-	-
PC6. perform basic post-processing like trimming or assembly	3	5	-	-
PC7. participate in fitment trials or ergonomic checks	3	5	-	-
PC8. report observed deviations and provide input for design iteration	2	4	-	-
<i>Assist in Functional Testing</i>	8	13	-	-
PC9. measure strength, stability, or deformation under stress	3	5	-	-
PC10. collect and organize testing data	2	3	-	-
PC11. identify visible or functional defects in prototypes	3	5	-	-
<i>Support Iteration and Documentation</i>	12	20	-	-
PC12. incorporate design changes based on test results	3	5	-	-
PC13. track design revisions and create test summaries	3	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. follow documentation formats for prototype evaluation	3	5	-	-
PC15. assist in preparing technical review reports	3	5	-	-
NOS Total	40	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4410
NOS Name	Assist in Prototyping and Product Testing
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Design, Plastic 3D Designing, Research and development, Manufacturing/process, Testing and Quality Control
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N8003.Assist in product design & development	30	70	-	-	100	20
RSC/N4409.Create and Modify 2D/3D Drawings using CAD Software	40	60	-	-	100	40
RSC/N4410.Assist in Prototyping and Product Testing	40	70	-	-	110	30
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	160	300	-	-	460	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.